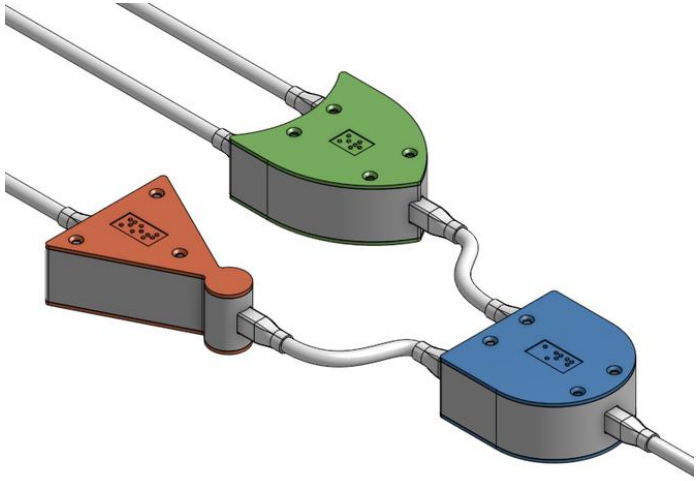
جامعة شقراء
Shaqra University



CNE 306 - Digital Logic Design
306 هحس - تصميم المنطق الرقمي

Second Term - 1446
Lecture #1

Chapter 2: Number Systems, Operations, and Codes

Decimal Numbers

- Position of each digit in a weighted number system is assigned a weight based on the base or radix of that system.
- Radix/base of decimal system is 10.
  - Ten possible values for each digit: 0, 1, 2, …, 9.
  - Example: The number 409.631_{10}
- Column weights of decimal numbers are powers of 10:

$$\ldots\ 10^3\ 10^2\ 10^1\ 10^0.\ 10^{-1}\ 10^{-2}\ 10^{-3}\ 10^{-4}\ \ldots$$

Radix Point (Decimal Point)

Decimal Numbers (Cont.)

- Decimal numbers can be expressed as sum of products of each digit times column value for that digit.
- Example:
  - Express the number 480.52_{10} as the sum of values of each digit.
- Solution:
  - $480.52_{10} = (4 \times 10^2) + (8 \times 10^1) + (0 \times 10^0) + (5 \times 10^{-1}) + (2 \times 10^{-2})$

Binary Numbers

- Digital systems use binary number system.
- Radix/base of system is 2.
  - Two possible values for each binary digit (know as bit): 0 or 1.
  - Example: The number 10010.001_2
  - Note: Values 0 and 1 can be easily represented in hardware using 2 distinct voltage values: low (e.g., 0 volt) and high (e.g., 5 volt)!
- Column weights of decimal numbers are powers of 2:

$$\ldots\ 2^3\ 2^2\ 2^1\ 2^0.\ 2^{-1}\ 2^{-2}\ 2^{-3}\ 2^{-4}\ \ldots$$

Binary Point

Conversion: Binary ➔ Decimal

- Method: Sum of weights (Add column values of all of bits that are 1 and discard all of bits that are 0).
- Example: Convert 100101.01_2 to decimal.
- Solution: Write column weights ➔ Add weights that correspond to 1's.

2^5	2^4	2^3	2^2	2^1	2^0	.	2^{-1}	2^{-2}
32	16	8	4	2	1	.	½	¼
1	0	0	1	0	1	.	0	1
32			+4		+1			+¼

$= 37.25_{10}$

Conversion: Decimal ➔ Binary (Integer)

- Method #1: Reverse sum-of-weights (Write down column weights and place 1's in columns that sum to decimal num.)
- Example: Convert 49_{10} to binary.
- Solution: Write down column weights until the last number is larger than the one you want to convert.
  - 2^6 2^5 2^4 2^3 2^2 2^1 2^0
  - 64 32 16 8 4 2 1
  - 0 1 1 0 0 0 1 = 110001_2

Conversion: Decimal ➔ Binary (Integer)

- Method #2: Repeated division-by-2 (divide decimal num. by 2 until quotient is 0. Remainders form binary num.)
- Example: Convert 12_{10} to binary.
- Solution:

		Quotient	Remainder
12 ÷ 2	=	6	0
6 ÷ 2	=	3	0
3 ÷ 2	=	1	1
1 ÷ 2	=	0	1

STOP

1 1 0 0

Conversion: Decimal ➔ Binary (Fraction)

- Method: Repeated multiplication-by-2 (multiply fractional results repeatedly by 2. Carries form binary fraction)
- Example: Convert 0.3125_{10} to binary.
- Solution:

		Carry	Result
0.3125 * 2	=	0	.625
0.625 * 2	=	1	.25
0.25 * 2	=	0	.5
0.5 * 2	=	1	.0

0. 0 1 0 1

STOP

Counting in Binary

- Using 4 bits, we can represent numbers: 0 : 2^4-1
- Using n bits, we can represent numbers: 0 : 2^n-1

Decimal Number	Binary Number			
00	0	0	0	0
01	0	0	0	1
02	0	0	1	0
03	0	0	1	1
04	0	1	0	0
05	0	1	0	1
06	0	1	1	0
07	0	1	1	1
08	1	0	0	0
09	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

Hexadecimal Numbers

- Radix/base of system is 16.
  - Sixteen possible values for each digit: 0, 1, 2, …, 9, A, B, C, D, E, F.
  - Example: The number $3D57.F0E_{16}$

10_{10} 12_{10} 14_{10}

11_{10} 13_{10} 15_{10}

- Column weights of hexadecimal num's are powers of 16:

$$\ldots\ 16^3\ 16^2\ 16^1\ 16^0.\ 16^{-1}\ 16^{-2}\ 16^{-3}\ 16^{-4}\ \ldots$$

Conversion: Hexadecimal ➔ Decimal

- Method: Sum of weights
- Example: Convert $3FA.4_{16}$ to decimal.
- Solution:

16^2	16^1	16^0	.	16^{-1}		
256	16	1	.	1/16		
*	*	*		*		
3	F	A	.	4		
768	+240	+10		+0.25	=	1018.25_{10}

Conversion: Decimal ➔ Hexadecimal

- Method:
  - Repeated division-by-16 (for integer)
  - Repeated multiplication-by-16 (for fraction)
- Example: Convert 943_{10} to Hexadecimal.
- Solution:

	Quotient	Remainder
943 ÷ 16 =	58	15
58 ÷ 16 =	3	10
3 ÷ 16 =	0	3

STOP

3 A F

Conversion: Binary ➔ Hexadecimal

- Method: 4-bit grouping (Break binary num. into 4-bit groups starting from radix point; replace each 4-bit group with the equivalent hexadecimal symbol)
- Example: Convert 10011101010.101111_2 to hexadecimal.
- Solution:

0100 1110 1010 . 1011 1100

4 E A . B C = $4EA.BC_{16}$

NOTE: A zero added to the right of a fraction or the left of an integer, doesn't change its value!!

Conversion: Hexadecimal ➔ Binary

- Method: 4-bit replacement (Replace each hexadecimal symbol with the appropriate four bits)
- Example: Convert $2B9.3A_{16}$ to binary.
- Solution:

2 B 9 . 3 A =

0010 1011 1001 . 0011 1010 = 1010111001.0011101_2

NOTE: A zero removed from the right of a fraction or the left of an integer doesn't change its value!!

Octal Numbers

- Radix/base of system is 8.
  - Eight possible values for each digit: 0, 1, 2, …, 7.
  - Example: The number 2407.321_8
- Column weights of octal numbers are powers of 8:

$$\ldots\ 8^3\ 8^2\ 8^1\ 8^0.\ 8^{-1}\ 8^{-2}\ 8^{-3}\ 8^{-4}\ \ldots$$

Conversion: Octal ⇔ Dec./Hex./Bin.

- Methods are similar to hexadecimal; with few changes.
  - Octal → decimal: sum of weights
  - Decimal → octal: repeated division-by-8 and repeated multiplication-by-8
  - Binary → octal: 3-bit grouping
  - Octal → binary: 3-bit replacement

Conversion Methods Summary

From \ To	Decimal	Binary	Octal	Hexadecimal
Decimal	-	Repeated Div./Mult. By 2	Repeated Div./Mult. By 8	Repeated Div./Mult. By 16
Binary	Sum of Weights	-	3-bit Grouping	4-bit Grouping
Octal	Sum of Weights	3-bit Replacement	-	Oct. ➔ Bin. Bin. ➔ Hex.
Hexadecimal	Sum of Weights	4-bit Replacement	Hex. ➔ Bin. Bin. ➔ Oct.	-

Unsigned Binary Addition

- Eight basic rules for adding binary digits (bits):

When there is no carry (or a carry of 0)	When there is a carry of 1
0 + 0 ➔ Sum = 0, carry = 0	1 + 0 + 0 ➔ Sum = 1, carry = 0
0 + 1 ➔ Sum = 1, carry = 0	1 + 1 + 0 = 10 ➔ Sum = 0, carry = 1
1 + 0 ➔ Sum = 1, carry = 0	1 + 0 + 1 = 10 ➔ Sum = 0, carry = 1
1 + 1 ➔ Sum = 0, carry = 1	1 + 1 + 1 ➔ Sum = 1, carry = 1

- Example:

$$
\begin{array}{cccccc}
 & 1 & 1 & 1 & 0 & \\
 & 0 & 1 & 1 & 1 & 0 \\
+ & 0 & 0 & 1 & 1 & 1 \\
\hline
 & 1 & 0 & 1 & 0 & 1
\end{array}
$$

Unsigned Binary Subtraction

- Eight basic rules for subtracting binary digits (bits):

When there is no borrow (or a borrow of 0)	When there is a borrow of 1
0 – 0 ➔ sum = 0, borrow = 0	0 – 0 – 1 ➔ sum = 1, borrow = 1
0 – 1 ➔ sum = 1, borrow = 1	0 – 1 – 1 ➔ sum = 0, borrow = 1
1 – 0 ➔ sum = 1, borrow = 0	1 – 0 – 1 ➔ sum = 0, borrow = 0
1 – 1 ➔ sum = 0, borrow = 0	1 – 1 – 1 ➔ sum = 1, borrow = 1

- Example:

$$
\begin{array}{rccccc}
 & 1 & 1 & 1 & 0 & \\
 & 1 & 0 & 1 & 0 & 1 \\
- & 0 & 1 & 1 & 1 & 1 \\
\hline
 & 0 & 0 & 1 & 1 & 0
\end{array}
$$

Unsigned Binary Multiplication

- Four basic rules for multiplying binary digits (bits):
  - $0 \times 0 = 0,\ 0 \times 1 = 0,\ 1 \times 0 = 0,\ 1 \times 1 = 1$
- Example:

					1	0	1	1	Multiplicand
				×	1	1	0	1	Multiplier
					1	0	1	1	
				+	0	0	0		Partial Products
			+	1	0	1	1		
		+	1	0	1	1			
1	0	0	0	1	1	1	1		Product

Unsigned Binary Division

- Same procedure as decimal division ➔ Long Division
- Example:

					0	0	0	0	1	1	0	1			Quotient
Divisor	1	0	1	1	1	0	0	1	0	0	1	1			Dividend
						1	0	1	1						
					0	0	1	1	1	0					
							1	0	1	1					
							0	0	1	1	1	1			
									1	0	1	1			
									0	1	0	0			Remainder

Partial Remainders

Reading Material & Exercises

- Floyd, Chapter 2:
  - Pages 67 – 77
  - Pages 92 – 95
  - Pages 98 – 100
  - Problems 1 – 18
  - Problems 37 – 40
  - Problems 43 – 46

Section 2–2 Binary Numbers:

7. Convert each binary number to decimal:

(g) 1011010.1010

8. What is the highest decimal number that can be represented by each of the following numbers of binary digits (bits)?

(h) nine

9. How many bits are required to represent the following decimal numbers?

(g) 132

10. Generate the binary sequence for each decimal sequence:

(e) 64 through 75

Section 2–3 Decimal-to-Binary Conversion:

11. Convert each decimal number to binary by using the sum-of-weights method:

(g) 125

12. Convert each decimal fraction to binary using the sum-of-weights method:

(b) 0.246

13. Convert each decimal number to binary using repeated division by 2:

(f) 59

14. Convert each decimal fraction to binary using repeated multiplication by 2:

(b) 0.347

Section 2–4 Binary Arithmetic:

15. Add the binary numbers:

(e) 1001 + 101

16. Use direct subtraction on the following binary numbers:

(f) 11010 – 10111

17. Perform the following binary multiplications:

(d) 1001 × 110

18. Divide the binary numbers as indicated:

(b) 1001 ÷ 11

Section 2–8 Hexadecimal Numbers:

37. Convert each hexadecimal number to binary:

(d) $5C8_{16}$

40. Convert each decimal number to hexadecimal:

(f) 2890

42. Perform the following subtractions:

(b) $C8_{16} - 3A_{16}$

Section 2–9 Octal Numbers:

43. Convert each octal number to decimal:

(g) 163_8

46. Convert each binary number to octal:

(h) 10110000011